BIRZEIT UNIVERSITY

Electrical and Computer Engineering

ENCS4310 |Digital Signal Processing| Semester 2 / 2025-2026|Assignment 1

Correlation

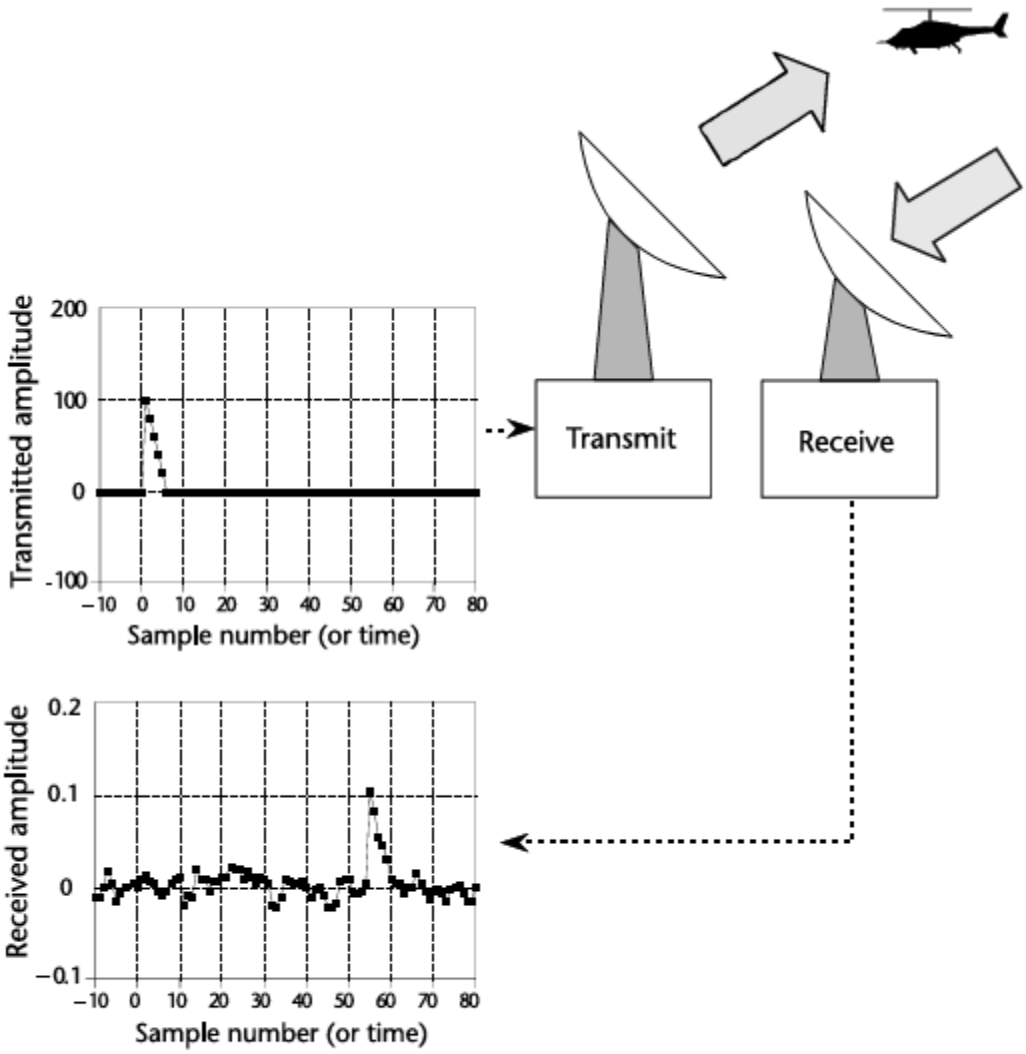


Figure Key elements of a radar system. Like other echo location systems, radar transmits a short pulse of energy that is reflected by objects being examined. This makes the received waveform a shifted version of the transmitted waveform, plus random noise. Detection of a known waveform in a noisy signal is the fundamental problem in echo location. The answer to this problem is *correlation*.

Cross-correlation and autocorrelation are two important concepts in DSP. Cross-correlation is a measure of similarity of two series as a function of the displacement of one relative to the other (delay):

$$R_{x,y}(k) = \sum_{-\infty}^{\infty} x[n]y[n+k]$$

where k is the displacement.

Auto correlation is a special case:

$$R_{x,x}(k) = \sum_{-\infty}^{\infty} x[n]x[n+k]$$

Task1: What is the relation between convolution and correlation? (Discuss the expression and the use)

Discuss:

- Mathematical relation
- Key difference
- Typical applications

Similarity Detection Problem

Correlation is a measure of how similar signals are:

For time-limited sequences

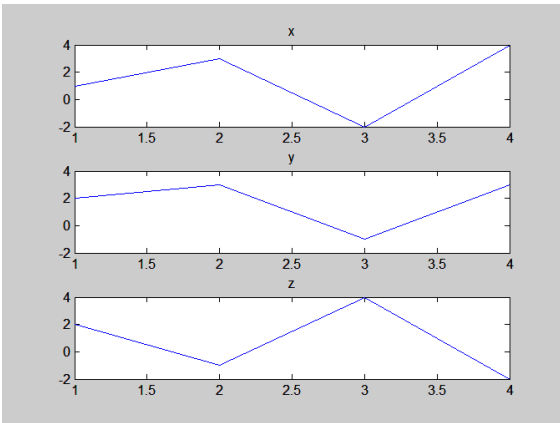
$$R_{x,y}(0) = \sum_{n=0}^{N-1} x[n]y[n]$$

Example:

Let x= [1 3 -2 4]

y= [2 3 -1 3]

z= [2 -1 4 -2]



$$R_{x,y} = 25$$

$$R_{x,z} = -17$$

It is clear that **y** is very similar to **x** than **z** snice $R_{x,y} > R_{x,z}$.

What about the following case:

If we modify z to w= [100 -1 4 -2]

- Repeat correlation with x
- Explain why correlation gives misleading results

We turn into **normalized correlation** (correlation coefficient) instead which is defined as:

$$\sigma_{x,y} = \frac{\sum_{n=0}^{N-1} x[n]y[n]}{\sqrt{\sum_{n=0}^{N-1} x^2[n]}\sqrt{\sum_{n=0}^{N-1} y^2[n]}}$$

- Compute normalized correlation between:

- x & y
- x & z
- x & w

Compare results and explain:

- Why normalization improves detection
- When normalized correlation should be used

Task2: check which one (standard or normalized) is more beneficial to identify how strongly one signal is present in two different signals. (Matlab)

Let Si= cos(2*π*fi*t)

s_1: 1 Hz s_2: 4 Hz s_3: 10 Hz

The following signals contain the three sinusoids above

X = 2S1+ 4S2 + S3

Y = S1+ S2

a) Plot the generated signals for 1 sec. let the sampling interval 0.01 sec

b) Compute:
   - Standard correlation between X and s_1
   - Normalized correlation between X and s_1

c) Repeat for Y and X and s_1

d) Which method better reflects the **strength of presence of** s_1?

e) Why does standard correlation depend on amplitude?

f) Which method would you use in:

   - Signal detection?
   - Pattern recognition?

More tasks will be given later!